8.4.14

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October 11,2025

Question

Let $\mathbf{P}(x_1, y_1)$ and $\mathbf{Q}(x_2, y_2)$, $y_1 < 0, y_2 < 0$, be the endpoints of the latus rectum of the ellipse $x^2 + 4y^2 = 4$ Then, the equations of the parabolas with latus rectum PQ are

1. $x^2 + 2\sqrt{3}y = 3 + \sqrt{3}$
2. $x^2 - 2\sqrt{3}y = 3 + \sqrt{3}$
3. $x^2 + 2\sqrt{3}y = 3 - \sqrt{3}$
4. $x^2 - 2\sqrt{3}y = 3 - \sqrt{3}$

Solution

Consider the ellipse defined by the quadratic form

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0, \tag{1}$$

where

$$\mathbf{V} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}, \quad \mathbf{u} = \mathbf{0}, \quad f = -4. \tag{2}$$

The eigenvalues and eigenvectors of $\mathbf{V}$ are

$$\lambda_1 = 1, \quad \mathbf{p}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \lambda_2 = 4, \quad \mathbf{p}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{3}$$

The eccentricity e of the ellipse is given by

$$e = \sqrt{1 - \frac{\lambda_1}{\lambda_2}} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}. \tag{4}$$

Solution

Define the vector

$$\mathbf{n} = \sqrt{\lambda_2}\mathbf{p}_1 = 2\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}2\\0\end{bmatrix}. \tag{5}$$

The formula for c is:

$$c = \frac{\mathbf{u}^\top\mathbf{n} \pm \sqrt{e^2(\mathbf{u}^\top\mathbf{n})^2 - \lambda_2(e^2-1)(\|\mathbf{u}\|^2 - \lambda_2 f)}}{\lambda_2 e(e^2-1)}. \tag{6}$$

Since $\mathbf{u} = \mathbf{0}$, this simplifies to:

$$c = \pm\frac{\sqrt{-\lambda_2(e^2-1)(-\lambda_2 f)}}{\lambda_2 e(e^2-1)}. \tag{7}$$

Solution

Calculate the terms inside the square root:

$$e^2 - 1 = \frac{3}{4} - 1 = -\frac{1}{4}, \tag{8}$$

$$-\lambda_2(e^2 - 1)(-\lambda_2 f) = -4 \times \left(-\frac{1}{4}\right) \times (-4 \times -4) = 16. \tag{9}$$

Thus,

$$c = \pm\frac{\sqrt{16}}{4 \times \frac{\sqrt{3}}{2} \times \left(-\frac{1}{4}\right)} = \mp\frac{2}{\sqrt{3}} \tag{10}$$

The focus **F** of the ellipse can be calculated as

$$\mathbf{F} = \frac{ce^2\mathbf{n}}{\lambda_2} = \frac{c \times \frac{3}{4} \times \begin{bmatrix} 2 \\ 0 \end{bmatrix}}{4} = \frac{3c}{8}\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \frac{3c}{4}\begin{bmatrix} 1 \\ 0 \end{bmatrix}. \tag{11}$$

Solution

Choosing $c = \frac{2}{\sqrt{3}}$, we get

$$\mathbf{F} = \frac{\sqrt{3}}{2}\begin{bmatrix}1\\0\end{bmatrix}. \tag{12}$$

The endpoints $\mathbf{x}_1, \mathbf{x}_2$ of the latus rectum PQ, where $y_1, y_2 < 0$, satisfy the ellipse equation

$$x^2 + 4y^2 = 4. \tag{13}$$

Substituting $x = \sqrt{3}$, we find

$$y = -\frac{1}{2}, \tag{14}$$

so the endpoints are

$$\mathbf{P} = \begin{bmatrix}\sqrt{3}\\-\frac{1}{2}\end{bmatrix}, \quad \mathbf{Q} = \begin{bmatrix}-\sqrt{3}\\-\frac{1}{2}\end{bmatrix} \tag{15}$$

Solution

The midpoint (vertex) **V** of the latus rectum is

$$\mathbf{V} = \frac{\mathbf{P}+\mathbf{Q}}{2} = \begin{bmatrix} 0 \\ -\frac{1}{2} \end{bmatrix}. \tag{16}$$

The length of the latus rectum is

$$|\mathbf{P}-\mathbf{Q}| = 2\sqrt{3} = 4a, \tag{17}$$

which gives

$$a = \pm\frac{\sqrt{3}}{2}. \tag{18}$$

The parabola with vertex **V** and parameter a has equation:

$$x^2 = 4a(y - y_0), \tag{19}$$

Solution

where $y_0 = -\frac{1}{2}$. Substituting $a = \pm\frac{\sqrt{3}}{2}$, we get

Case 1: $a = +\dfrac{\sqrt{3}}{2}$

$$x^2 - 2\sqrt{3}y = 3 + \sqrt{3}. \tag{20}$$

Case 2: $a = -\dfrac{\sqrt{3}}{2}$

$$x^2 + 2\sqrt{3}y = 3 - \sqrt{3}. \tag{21}$$

Hence, the equations of the required parabolas are

$$\boxed{x^2 - 2\sqrt{3}y = 3 + \sqrt{3} \quad \text{and} \quad x^2 + 2\sqrt{3}y = 3 - \sqrt{3}.} \tag{22}$$

Code (C)

Part 1

```c
#include <stdio.h>
#include <math.h>

int main() {
    // Ellipse: x^2 + 4y^2 = 4
    // Latus rectum endpoints y < 0
    double x1 = sqrt(3.0);
    double x2 = -sqrt(3.0);
    double y1 = -0.5;
    double y2 = -0.5;

    // Midpoint (vertex of parabola)
    double vx = (x1 + x2) / 2.0;
    double vy = (y1 + y2) / 2.0;
```

Code (C)

Part 2

```c
    // Length of latus rectum
    double L = x1 - x2;
    double a = L / 4.0;

    // Case 1: a positive
    double f1 = -4 * a * vy; // move vy to RHS
    printf("Equation of parabola 1:\n");
    printf("x^2 - %.3lf*y = %.3lf\n", 4*a, f1);

    // Case 2: a negative
    a = -a;
    double f2 = -4 * a * vy;
    printf("Equation of parabola 2:\n");
    printf("x^2 - %.3lf*y = %.3lf\n", 4*a, f2);

    return 0;
}
```

Python Code

Part 1

```python
import ctypes
import numpy as np
import matplotlib.pyplot as plt

libm = ctypes.CDLL('code14.dll')
libm.sqrt.argtypes = [ctypes.c_double]
libm.sqrt.restype = ctypes.c_double

def c_sqrt(x):
    return libm.sqrt(x)

# Ellipse parameters
a_ellipse = 2.0
b_ellipse = 1.0

e = c_sqrt(1 - (b_ellipse**2) / (a_ellipse**2))
```

Python Code

Part 2

```
c = a_ellipse * e
# Latus rectum endpoints (y < 0)
xP, yP = c, -0.5
xQ, yQ = -c, -0.5

xV = (xP + xQ) / 2
yV = (yP + yQ) / 2

lengthPQ = np.sqrt((xP - xQ)**2 + (yP - yQ)**2)
a_p = lengthPQ / 4
# Parabola coefficient 2 * sqrt(3)
coeff = 2 * np.sqrt(3)

# Constant in parabola equation x^2 - 23 y = constant
constant = -coeff * yV
print(f"Parabola equation: x^2 - 23 y = {constant:.4f}")
```

Python Code: Plotting

Part 3

```
theta = np.linspace(0, 2*np.pi, 400)
x_ellipse = a_ellipse * np.cos(theta)
y_ellipse = b_ellipse * np.sin(theta)

# Parabola: x^2 = 23 (y + 0.5)
# => y = (x^2) / (23) - 0.5
x_parabola = np.linspace(-4, 4, 400)
y_parabola = (x_parabola**2) / (2 * np.sqrt(3)) - 0.5

plt.figure(figsize=(8, 6))
plt.plot(x_ellipse, y_ellipse, label='Ellipse: $x^2 + 4y^2 = 4$',
     color='blue')
plt.plot(x_parabola, y_parabola, label='Parabola', color='red')
# Mark latus rectum endpoints and vertex
plt.scatter([xP, xQ], [yP, yQ], color='green', label='Latus
    Rectum Endpoints P, Q')
plt.scatter(xV, yV, color='purple', label='Vertex V')
```

Python Code: Plotting

Part 4

```
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)

plt.title('Ellipse and Parabola with Latus Rectum')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.axis('equal')

# Save the plot
plt.savefig('fig14.png', dpi=300)
plt.show()
```

Graphical Representation